COMP5583 Agentic Computing, Systems, and Applications

Assignment: Personal Website Development with AI Agents

Individual assignment | Total: 100 marks

Deadline: 31 October 2026 at 23:59 (Hong Kong time, UTC+8)

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For any assignment questions, contact the TA and include COMP5583 in the email subject.

This assignment is designed to let you apply what you've learned in class to build and improve your own personal website with an LLM agent! We will do this in 3 steps, and each step will be a task:

- Task 1: Build a personal website with an LLM-based AI agent
- Task 2: Create your evaluator to evaluate your website.
- Task 3: Improve your website with 2 techniques and validate the improvement with your own evaluator.

You may use any LLMs available to you. There is no requirement to use paid models. Please specify the model(s) you used in your report.

Task 1. Build the Initial Website - 30 marks

Create a personal website with the following four sections. A single-page site with section links is acceptable.

Section	Required content
Home	A display name, a short introduction, and navigation to the other sections.
About	Your educational background and interests.
Projects	At least two projects, with their purposes and your contributions. Coursework and clearly labelled proposed projects are acceptable.
Contact	A suitable contact option, or a statement that contact details are withheld.

To include the content above, you can ask an LLM to help you with coding. A good place to host your website would be GitHub Pages. To make your website shinier, you can try adding interactive features such as a theme switch or an expandable project description.

Write your system prompt that gives the agent general instructions for building your website. If your tool has no system-prompt field, use an initial instruction prompt. Save the first website that meets the minimum content and functionality requirements as V0, together with the prompts you tested and relevant agent outputs. Essential repairs are allowed before freezing V0 and must be recorded. Do not apply the Task 3 techniques before freezing V0. Do not deliberately weaken the baseline.

Briefly explain what visitors could do on your website and how the prompt specifies the required LLM behaviour. All three website versions must meet the same requirements.

You may use a coding agent or a browser-based LLM in a human-supervised workflow. Paid tools, full automation, and a backend service are not required.

Mark allocation: content and functionality, 10; initial prompt, 10; visual design and usability, 10.

Task 2. Design and Use an Independent Evaluator - 30 marks

Brainstorm with an AI agent to design 4 criteria to form your own evaluator. Score each criterion from 0 to 5 and make sure their weights total 100%. You need to define each criterion, its scoring rules and the reason for using it in the report. The scoring rules should define what scores of 0, 3, and 5 mean.

Next, use the criteria to evaluate V0 yourself. List the score for each criterion and, more importantly, 1-2 examples that support the score. In your report, provide the details of each example: 1) the prompts or tests you used; 2) the relevant agent outputs or website behaviour; 3) the reason why they support the score. Manual self-scoring is required only for V0.

You can implement your evaluator as a reusable AI judging prompt, a script, or both. For an AI judge, use a fresh conversation containing only the fixed rubric, scoring instructions, and website evidence, without your self-scores or desired conclusions. The same underlying LLM may be used. Provide website evidence that the evaluator can inspect, such as screenshots, source files, or test results. Apply it to V0 and compare its scores with yours.

Judging by the score, what methods could you implement to address the weaknesses? Think about it and the answer will guide you in Task 3.

Mark allocation: rubric design, 10; experimental design and execution, 10; results and analysis, 10.

Task 3. Improve the Website Using Two Techniques - 40 marks

Based on the evaluation of V0, introduce at least two distinct techniques beyond the baseline workflow. Choose from the following examples or justify an alternative.

Technique	How it may be used
In-context learning	Add examples to your prompt for few-shot learning. Show how these examples influence the revised website.
Retrieval-augmented generation (RAG)	Retrieve relevant information from a small collection and supply it to the agent. Record the query, selected passages, and how they support the revision.

Other course techniques or relevant methods from papers published in 2022-2026 and available when this assignment is issued are acceptable. Cite the source and describe any changes you made to the published method.

Apply the first and second techniques separately to V0 to get V1 and V2, respectively. V1 starts from V0 with Technique 1; V2 starts from V0 with Technique 2, not from V1. Deploy V0, V1 and V2 as three separate webpages.

Next, use the same evaluator to evaluate V1 and V2. Record the same information as for V0 and then analyze the differences between V0, V1 and V2. Do your techniques make an improvement or not? Why? Write it in your report.

Use a suitable test, screenshot comparison, or relevant evaluation score to support each claim. A change in an overall AI score alone is not enough to establish why the website changed. Explain any trade-offs, failed attempts, or remaining problems.

Two minor prompt rewrites or routine execution repairs do not count as two techniques. Identify two techniques for marking; additional techniques do not increase the available marks.

Mark allocation: 10 marks for implementing each technique and 10 marks for testing and explaining its effect. Two techniques give 40 marks in total.

Submission

Submit COMP5583_StudentID.zip by **31 October 2026** at 23:59 (Hong Kong time, UTC+8), replacing StudentID with your student number. Include all four items below.

Use a PDF main report of no more than eight pages: A4, 12-point Arial, and single line spacing. The limit includes the title page, figures, tables, and references. Place your name, student ID, and all three working URLs, labelled V0, V1 and V2, on the first page.

Follow Tasks 1-3 in the report. Keep the analysis in the main report; the required prompt appendix has no page limit. All three webpages must open without a login and stay available until marking is complete. Keep V0 unchanged.

File or folder	Required contents
report.pdf	Main report, maximum eight pages. Include V0, V1 and V2 URLs on the first page.
website/	Separate V0, V1 and V2 source folders with assets and a short README explaining how to open or run them.
prompt_appendix.pdf	Required, no page limit. All prompts you supplied for Tasks 1-3, relevant outputs, retrieval records, and test feedback, labelled by task and version.
evaluation/	Scoring prompt or script, scoring rules, test evidence, and saved evaluation results.

You may use a pseudonym on public pages. Do not publish student IDs, private personal information, or API keys. Credit external text, images, templates, and methods.

Marking Rubric

Total: 100 marks. The following rules apply to the three tasks. For each 10-mark criterion: 8-10 means substantially complete and well supported; 5-7 means partly complete with material gaps; 1-4 means limited or weakly supported work; 0 means no relevant assessable work. Apply these bands to the stated basis for marking.

Task 1. Website and Initial Prompt (V0) - 30 marks

Criterion	Marks	Basis for marking
Content and functionality	10	All four sections contain the required information.
Instruction prompt	10	The prompt clearly specifies the website requirements and supplied content; the baseline and essential repairs are recorded.
Visual design and usability	10	The V0 website is coherent and readable, with appropriate spacing, consistent styling, and usable navigation.

Task 2. Independent Evaluation - 30 marks

Criterion	Marks	Basis for marking
Rubric design	10	Four criteria match the stated goal. Weights and 0-5 scoring rules are clear and usable.
Experimental design and execution	10	The evaluator scores V0 using the defined rubric.
Results and analysis	10	Criterion scores and totals are reported correctly. Comparisons, verification of two evaluator findings, and discussion of limitations support the conclusions.

Task 3. Techniques and Evidence - 40 marks

Criterion	Marks	Basis for marking
Technique 1: implementation	10	The technique is distinct, appropriate to the stated problem, and correctly applied. Prompts and relevant inputs show how it was used.
Technique 1: effectiveness evidence	10	Compare V1 with V0 using the evaluator developed in Task 2, and support the conclusion with relevant evidence.
Technique 2: implementation	10	A second distinct technique is correctly applied and documented to the same standard.
Technique 2: effectiveness evidence	10	Compare V2 with V0 using the evaluator developed in Task 2, and support the conclusion with relevant evidence.

Marks reflect the quality of implementation and evidence, not the number of tools used or the size of an improvement. Partial marks are given where requirements or evidence are incomplete. A carefully tested method that does not improve the website may still receive full evidence and analysis marks.

Your evaluator's website scores are evidence for the report, not your assignment grade.